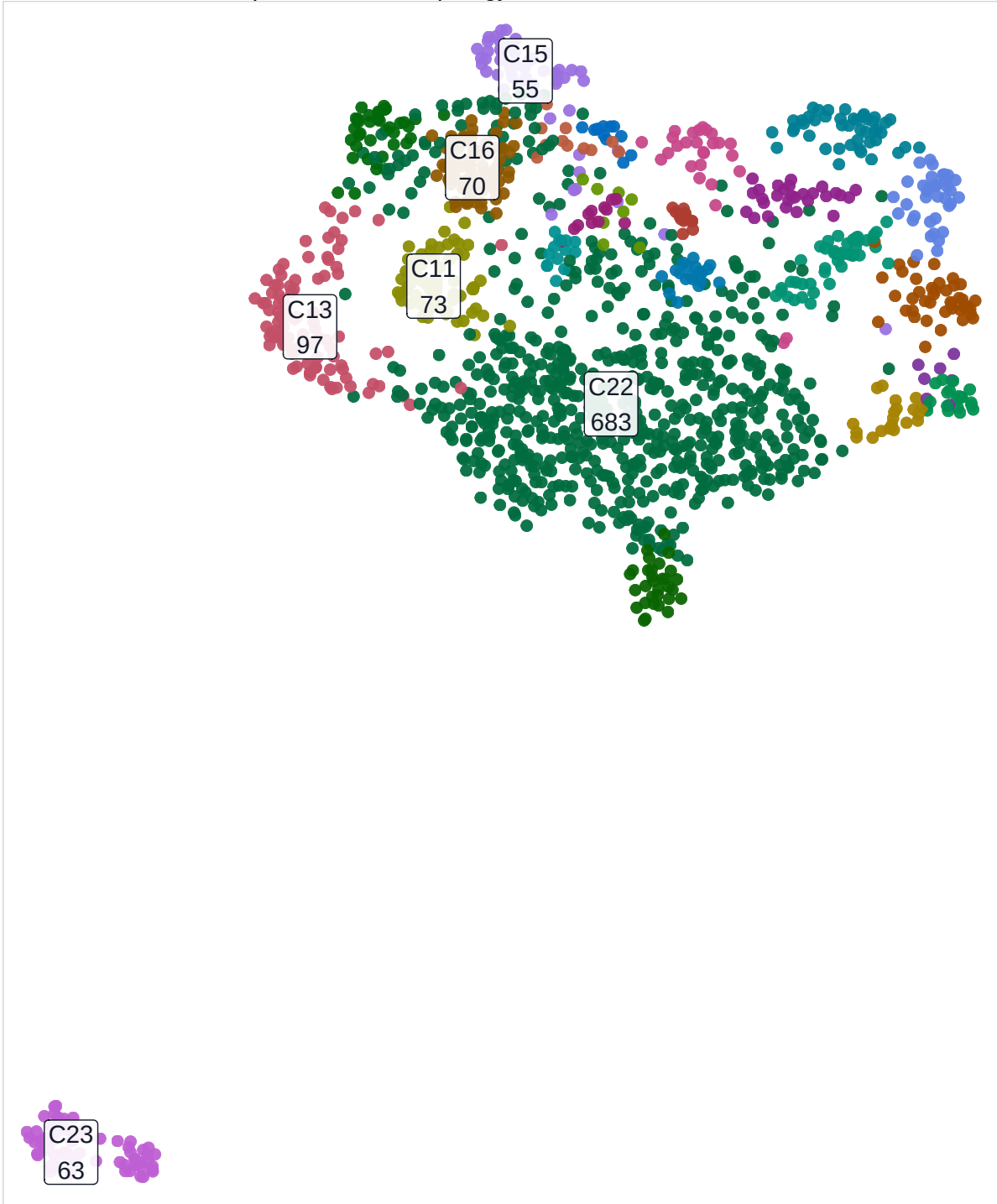


Goncalves cells: TBS clusters versus progenitor context

All four UMAPs use identical coordinates; only the coloring changes

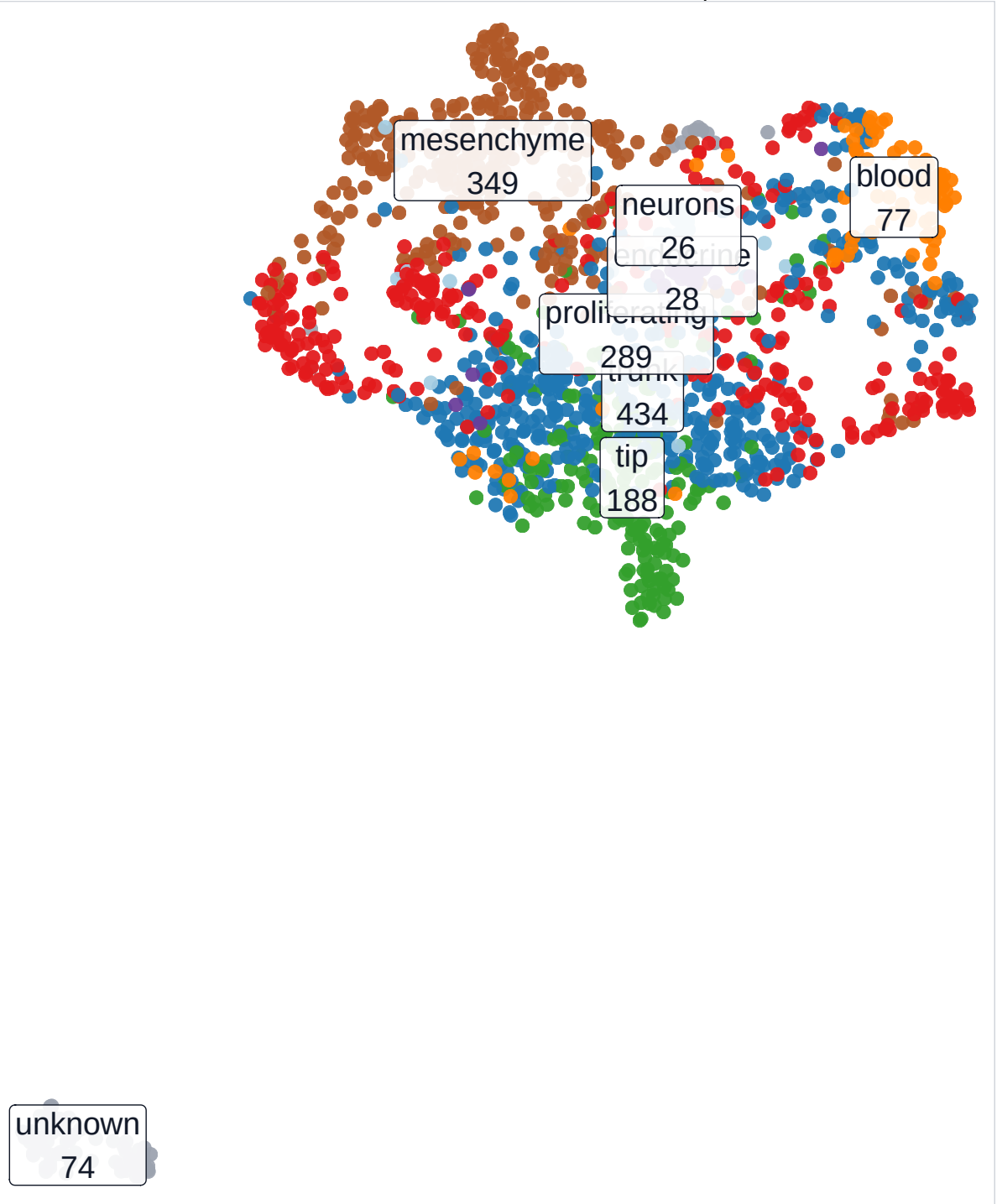
UMAP by TBS cluster

24 clusters from adaptive-diffusion topology TBS; labels shown for clusters >= 50 cells



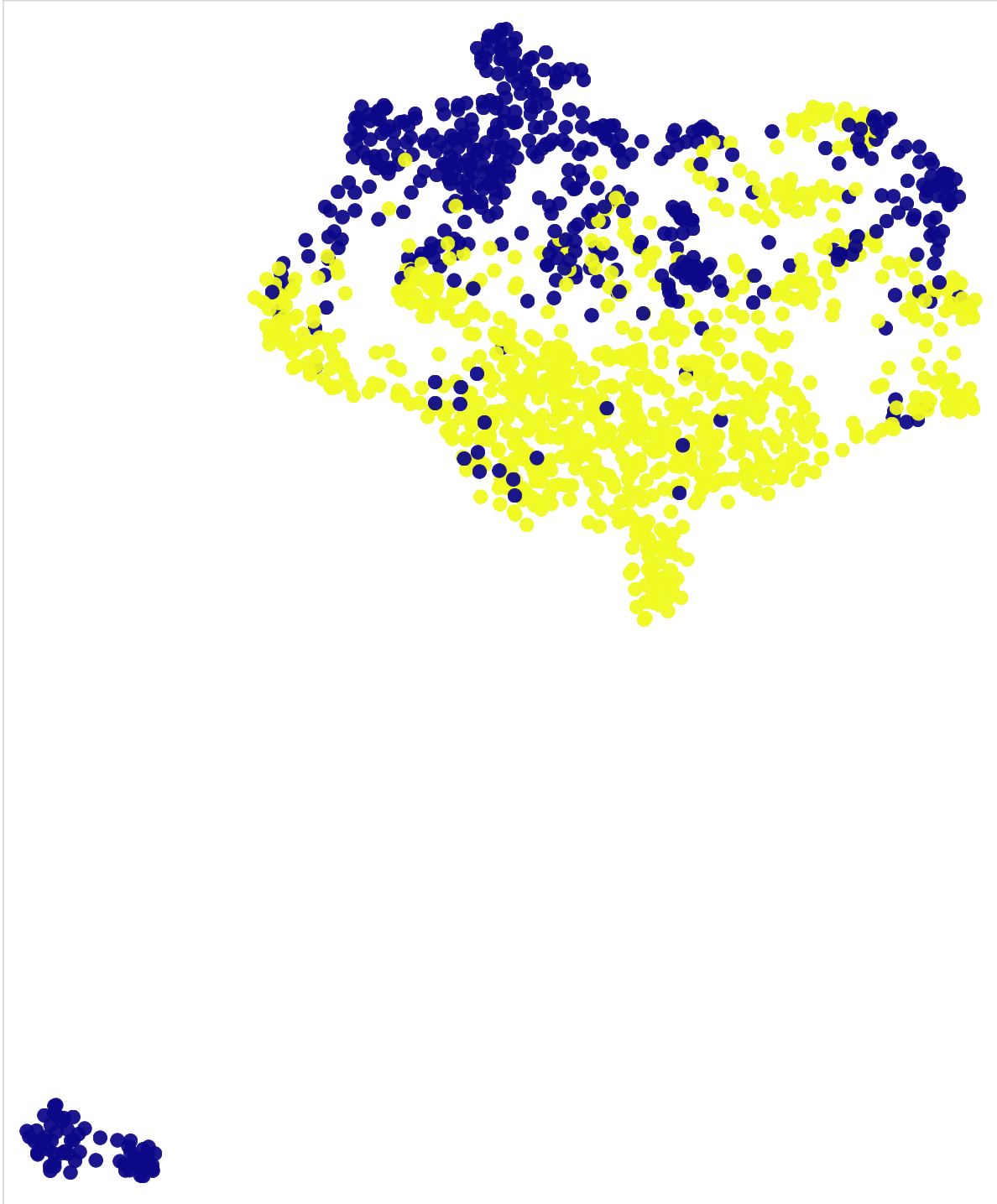
UMAP by fetal population

Same cell labels used for terminal tree tips



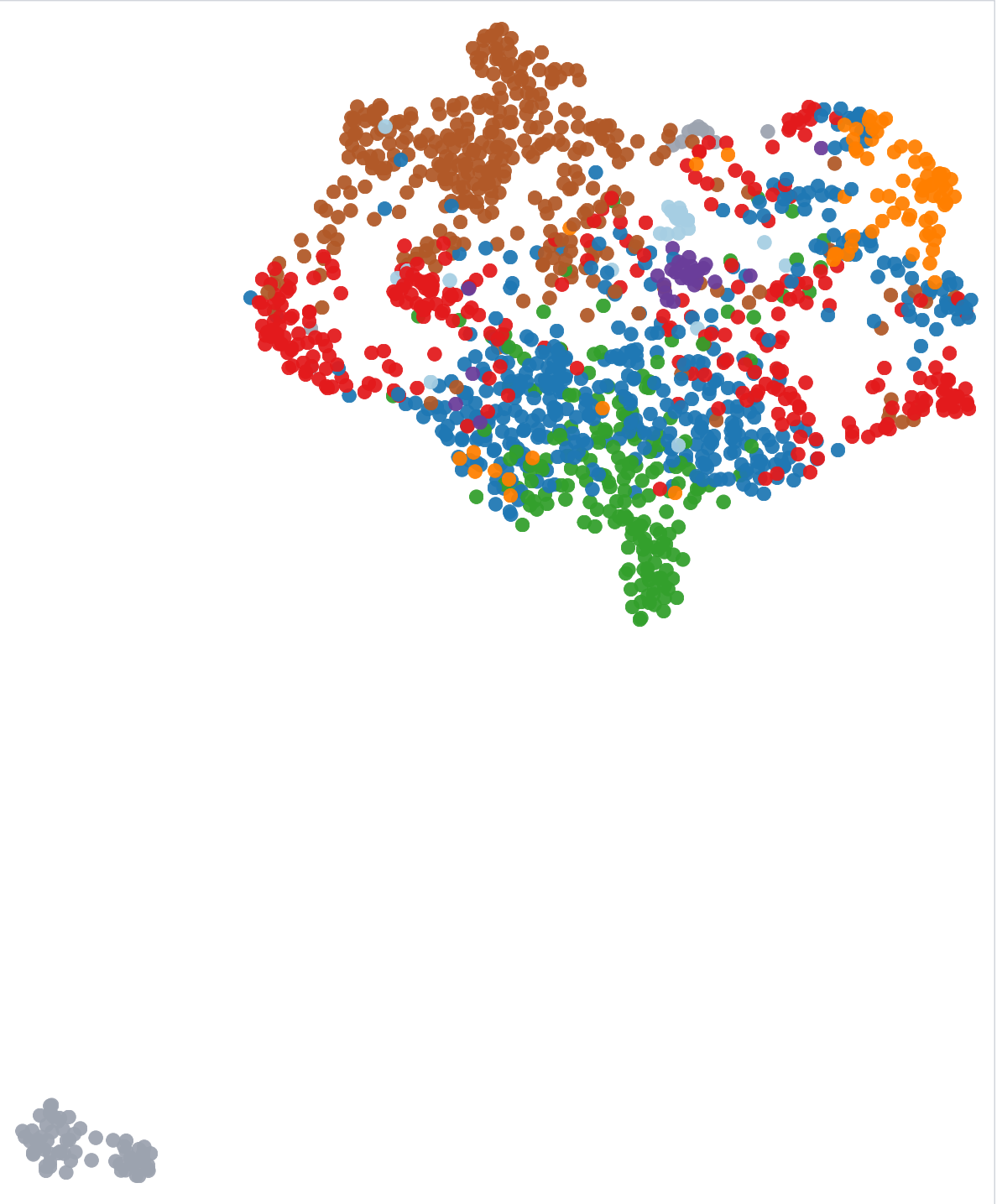
UMAP by progenitor-label membership

Tree color is descendant fraction; cell-level UMAP is 0/1 membership



UMAP by progenitor interpretation

Terminal cells use population-derived state classes



Cell is trunk/tip/prolif
0.00 0.25 0.50 0.75 1.00

tip progenitor partial progenitor tip cell blood other
proliferating progenitor endocrine proliferating cell neurons
mixed fetal progenitor trunk cell mesenchyme unknown